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Abstract

This report provides an overview of the information systems used by KFC, a global fast-food chain. The report starts with an introduction to KFC and its background. It then delves into the different types of information systems used by KFC, including their point of sale (POS) system, supply chain management system, and customer relationship management (CRM) system. The report also highlights the benefits of these information systems, such as increased efficiency, better inventory management, and improved customer experience. Additionally, the report discusses the challenges that KFC faces in implementing and maintaining these information systems. The conclusion summarizes the main findings of the report and provides recommendations for KFC to enhance their information systems.

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1. INTRODUCTION TO KFC & IT'S INFORMATION SYSTEMS

KFC (Kentucky Fried Chicken) is a global fast-food chain that is known for its fried chicken. It was founded in 1952 by Colonel Harland Sanders in Louisville, Kentucky, and has since grown to become one of the world's largest restaurant chains. Today, KFC has more than 24,000 locations in over 145 countries, serving millions of customers each day.



Figure 1 logo (KFC Organization)

Operational support systems are used to manage day-to-day business activities, such as sales transactions and inventory management, Management support systems are designed to support managerial decision-making and planning activities and knowledge support systems are designed to support the knowledge management activities of the organization.

KFC's information systems infrastructure plays a critical role in supporting its business operations and decision-making processes. By utilizing a variety of operational, management, and knowledge support systems, KFC can effectively manage its global operations, monitor performance, and provide a seamless customer experience.

To keep up with its vast operations and customer base, KFC utilizes a range of information systems to manage its business operations. These systems include point-of-sale (POS) systems, inventory management systems, supply chain management systems, and customer relationship management systems. By these systems, KFC is able to efficiently manage its operations, optimize its supply chain, monitor customer preferences, and provide a seamless experience to its customers. Additionally, KFC has also incorporated digital technologies to enhance its customer engagement and improve its operations.

2. DEFINITION OF INFORMATION SYSTEMS

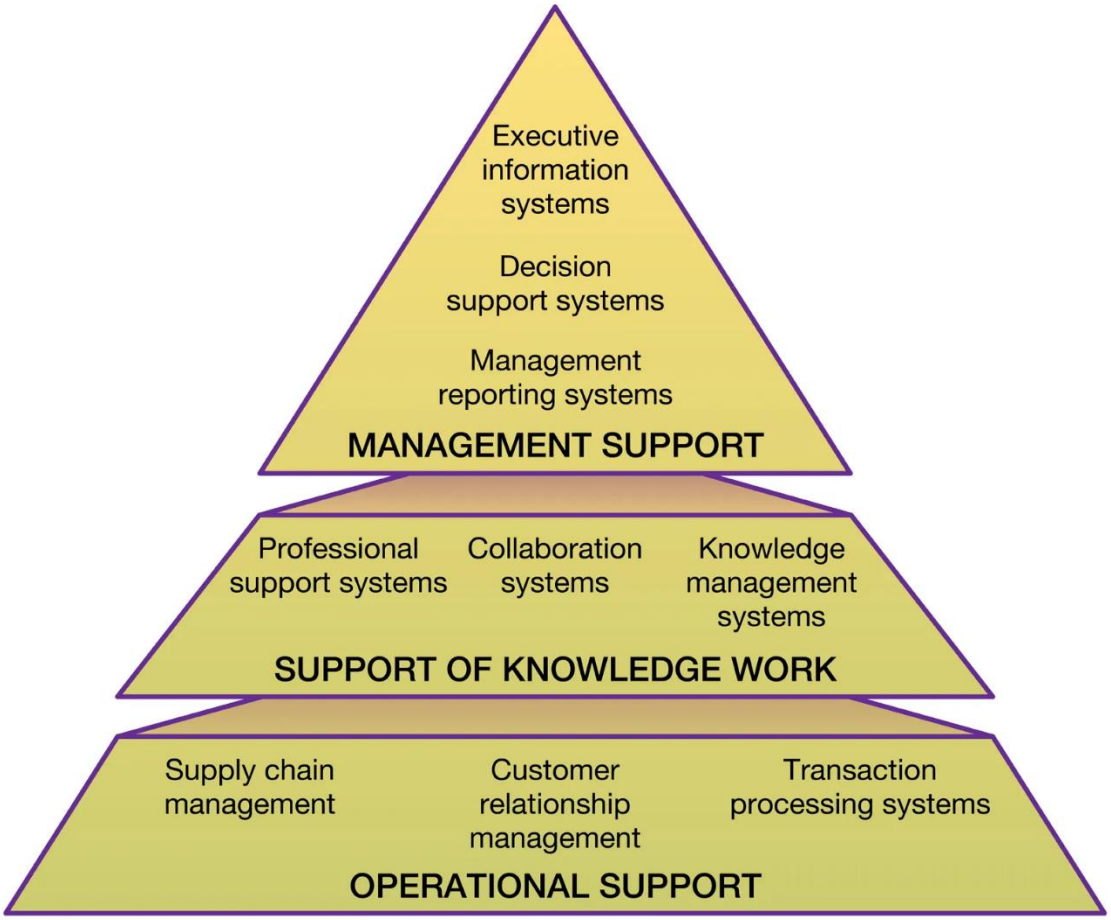


Figure 2 Information Systems (Encyclopedia Britannica 2012)

An organization's information systems consist of a network of interdependent parts that collaborate to gather, process, preserve, and distribute information and data. These systems use a combination of hardware, software, data, and people to support business operations and decision-making processes. Information systems are critical to the success of modern organizations, providing the necessary tools to manage and analyze vast amounts of data, streamline processes, and improve communication and collaboration across the organization. They play a key role in enabling businesses to adapt to changing market conditions, stay competitive, and make informed decisions. In recent years, advancements in technology have led to the development of new types of information systems, including cloud-based systems, mobile applications, and artificial intelligence-powered systems, further increasing their value to organizations of all sizes and industries.

2.1. Operation Support Information Systems

Operational Support Information System is a type of information system that supports day-to-day operations. OSIS provides information to support routine, repetitive, and structured decisions. Examples of Operational Support Information System used in KFC are:

- Point of Sale (POS) System
- Inventory Management System

2.1.1. Point of Sale (POS) System

A POS system is used to process transactions at the restaurant. At KFC, a POS system could be used to process orders, calculate taxes, and generate receipts.

2.1.2. Inventory Management System

An inventory management system is used to manage the inventory levels of raw materials, finished goods, and supplies. At KFC, an inventory management system could be used to track the inventory levels of chicken, buns, sauces, and other ingredients.

2.2. Knowledge Support Information Systems

Knowledge Support Information System is a type of information system that helps organizations create, store, and disseminate knowledge. It also allows employees to access the information they need to perform their jobs more efficiently. Examples of Knowledge Support Information System used in KFC are:

- Expert System

2.2.1. Expert Systems

This system is a computer program that takes off the decision-making ability of a human expert. At KFC, an expert system could be used to recommend the best menu items for customers based on their preferences and to store recipes, cooking techniques, and best practices that are used to prepare KFC's signature dishes.. This also helps organizations manage their knowledge assets.

2.3. Management Support Information System

Management Support Information System is a type of information system that provides information to top-level management to support strategic decision-making. It provides timely, accurate, and relevant information that can help executives and managers make informed decisions. Examples of Management Support Information System used in KFC are:

- Executive Dashboard
- Decision Support System

2.3.1. Executive Dashboard

An executive dashboard is a visual representation of key performance indicators used to monitor the overall performance of the organization. At KFC, an executive dashboard could display sales, revenue, customer satisfaction, and employee productivity.

2.3.2. Decision Support System

A Decision support system is a computer-based system that helps decision-makers analyze data and make informed decisions. At KFC, a DSS could be used to determine which products to launch in new markets, what prices to set, and which marketing strategies to use.

3. INFORMATION SYSTEMS & IT'S COMPONENTS

Hardware:

The hardware components of the POS system include a terminal, a cash drawer, a receipt printer, and a card reader. The terminal is the main device that allows employees to input orders and process payments. The cash drawer is used to store cash received from customers. The receipt printer is used to print receipts for customers. The card reader is used to read payment card information.



Figure 3 Hardware (KFC Russia, 5)201



Figure 4 Software (Anon, n.d)

Software:

The POS software is the application that runs on the terminal and processes customer orders and payments. The software includes a graphical user interface (GUI) that allows employees to input orders, apply discounts and coupons, and process payments. The software also integrates with the kitchen system, which sends order details to the kitchen staff.

People:

The people component of the POS system includes the employees who use the system, including cashiers, servers, and kitchen staff. Cashiers and servers use the system to input customer orders and process payments, while kitchen staff use the system to receive and fulfill customer orders.



Figure 5 People (Loopenews,2019)

Process:

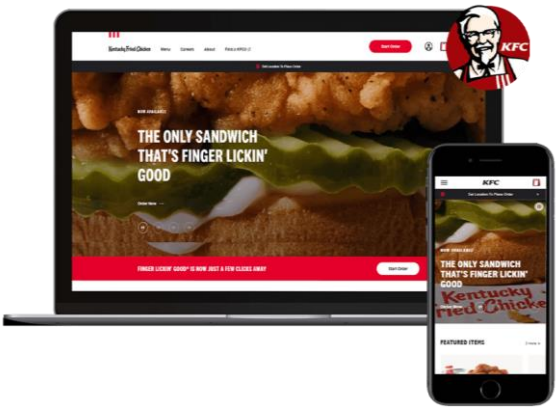


Figure 6 Process (Anon, n.d.)

The process component of the POS system includes the procedures employees follow when using the system. The process begins when a customer places an order, which is entered into the POS system by a cashier or server. The system then sends the order details to the kitchen staff, who prepare and cook the food. Once the food is ready, the kitchen staff mark the order as complete in the system, and the cashier or server retrieves the food and delivers it to the customer. Finally, the system processes the payment and generates a receipt for the customer.



Figure 7 Data (KFC Homepage, 2023)

Data:

The data component of the POS system includes customer order details, payment information, and sales data. The system records each customer order and stores the order details, including the items ordered, any discounts or coupons applied, and the payment method. The system also records sales data, such as the total number of orders processed, and the total amount of sales revenue generated. This data is used by KFC to analyze sales trends, forecast demand, and identify areas for improvement.

4. SECURITY THREATS AND MACHANISMS TO PREVENT THEM

4.1 Security Threats

There are several security threats that could potentially affect the information systems used by KFC. Here are a few examples:

- Point of sale system
- Inventory management system
- Human Resources Management System
- Customer Relationship Management System

4.1.1 Point of Sale (POS) System

The POS system is a prime target for cybercriminals who seek to steal payment card data. Attackers may attempt to install malware on the POS system to intercept payment card data as it is being processed. This can lead to significant financial losses and damage to KFC's reputation.

4.1.2 Inventory Management System

If the inventory management system is not properly secured, it may be vulnerable to unauthorized access or manipulation. Attackers could potentially modify inventory levels or ingredient lists, which could lead to incorrect orders or shortages.

4.1.3 Human Resources Management System

The human resources management system contains sensitive information about KFC's employees, including personal data and financial information. If this system is not properly secured, attackers could potentially gain access to this data and use it for identity theft or financial fraud.

4.1.4 Customer Relationship Management System

The customer relationship management system contains valuable customer data, such as contact information and purchasing habits. If this system is compromised, attackers could potentially steal this data and use it for targeted phishing or other types of fraud.

4.2 Security Mechanisms

Here are a few mechanisms that KFC can use to mitigate the security threats mentioned earlier:

- Point of sale system
- Inventory management system
- Human Resources Management System
- Customer Relationship Management System

4.2.1 Point of Sale (POS) System

To mitigate the threat of payment card data theft, KFC can implement end-to-end encryption to protect payment data as it is transmitted from the POS system to the payment processor. KFC can also implement two-factor authentication for access to the POS system, and regularly monitor the system for suspicious activity.

4.2.2 Inventory Management System

To prevent unauthorized access to the inventory management system, KFC can implement strong access controls, such as multi-factor authentication and role-based access control. KFC can also regularly review system logs to detect any suspicious activity or unauthorized access attempts.

4.2.3 Human Resources Management System

To prevent unauthorized access to the human resources management system, KFC can implement strong access controls, such as multi-factor authentication and role-based access control. KFC can also regularly review system logs to detect any suspicious activity or unauthorized access attempts. Additionally, KFC can implement data loss prevention technologies to prevent the unauthorized transfer of sensitive data.

4.2.4 Customer Relationship Management System

To protect the customer relationship management system, KFC can implement strong access controls and encryption for all data in transit and at rest. KFC can also regularly review system logs to detect any suspicious activity or unauthorized access attempts. Additionally, KFC can implement two-factor authentication for customer accounts and educate customers on how to protect their personal information.

5 Conclusion

In conclusion, KFC has implemented various types of information systems to support its operations, knowledge management, and management decision-making processes. The operational support information systems include the point of sale (POS) system, supply chain management system, and inventory management system. These systems help KFC to process orders, manage its suppliers, and monitor inventory levels efficiently. The knowledge support information systems include the customer relationship management (CRM) system, which helps KFC to gather and analyze customer data, and the knowledge management system, which enables KFC to store and share internal knowledge effectively.

Moreover, KFC has also implemented management support information systems, including the executive dashboard and decision support system. These systems provide management with real-time data and analytics, enabling them to make informed decisions quickly.

However, the implementation of these information systems exposes KFC to security threats, such as hacking, malware, and data breaches. To mitigate these threats, KFC must implement various security mechanisms, such as firewalls, encryption, and intrusion detection systems.

In summary, the implementation of information systems has enabled KFC to streamline its operations, improve knowledge management, and enhance decision-making processes. However, KFC must be vigilant about the security threats that these systems pose and implement various security mechanisms to safeguard its data and systems.

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7 Appendices

7.1. Workload matrix

Tasks	Akash 20222330	Juzail 20222336	Chadithya 20221798	Alfar 20221802	Dilina 20221803	Saad 20221804
Deciding a Company	X	X	X	X	X	X
Report						
Abstract						X
Acknowledgement						X
Introduction to KFC		X		X		
Introduction to IS			X	X		
IS types			X		X	
IS types used in KFC		X		X		
IS & it's components			X			
Security threats	X	X			X	
Mechanisms	X	X		X		
Conclusion						X
References						X
Meeting Minutes	X	X	X	X	X	X
Compile all info						X
Presentation					X	
Template					X	
Summarize the report	X	X	X	X	X	X
Animations			X			X

Table 1 Workload matrix

7.2. Meeting Agenda & Minutes

1st meeting minutes

Date : 17th February 2023
Time : 1.00 pm
Venue : Common room, IIT.
Attendees :

Akash Sathiyanathan
Mohamed Zabrin Mohamed Juzail
Chadithya Dinuwan
Mohammed Alfar
Dilina Chathuraka
Saad Mazhar

Time	Points
1.00 pm	○ Meeting was called to order at 1.00 pm. ○ The group discussed whom to appoint as a group leader and concluded that Saad Mazhar will be the leader for this module. ○ Dilina Chathuraka explained that a word document and a presentation should be prepared before the 15th of March 2023. ○ Chadithya Dinuwan asked the group to come up with company suggestion before the next meeting. ○ The group decided to communicate through WhatsApp and Zoom platforms, therefore, a WhatsApp group was created. ○ Mohamed Juzail mentioned that another meeting should be called within the week to finalize the company and start on the research. ○ The group agreed to conduct a virtual meeting within the week. ○ The meeting was adjourned at 1.30 pm.

Table 2 1st meeting

2nd Meeting Minutes

Date: 24th February 2023
Time : 4.00 pm
Venue : Cafeteria IIT
Attendees :

Akash Sathiyanathan
Mohamed Zabrin Mohamed Juzail
Chadithya Dinuwan
Mohammed Alfar
Dilina Chathuraka
Saad Mazhar

Time	Points
4.00 pm	○ Meeting was called to order at 4.00pm. ○ Every group member decided to pick any part from the report, which should be sent before the 1st of March. ○ Akash Sathiyanadan mentioned that he will be looking in the security threats and the mechanisms and submit the information on or before the 1st of March. ○ Dilina Chathuraka & Chadithya Dinuwan mentioned that they will be researching about Microsoft and its information systems as a whole. ○ The team decided to meet on the 1st of March with information in hand. ○ Meeting was adjourned.

Table 3 2nd meeting

3rd Meeting Minutes

Date : 28th February 2023
Time : 8.00 pm
Venue : Virtual Platform (Zoom)
Attendees :

Akash Sathiyanathan
Mohamed Zabrin Mohamed Juzail
Chadithya Dinuwan
Mohammed Alfar
Dilina Chathuraka
Saad Mazhar

Time	Points
8.00 pm	○ Meeting was called to order at 8.00 pm. ○ This emergency meeting was called to discuss if the group should proceed with Microsoft or switch. ○ A vote was called the majority decided to change the company because Microsoft was too common. ○ Few more company suggestions were brought forward, and the group decided to move forward with KFC. ○ Each member was told to research about KFC and put them forward in the next meeting. ○ The meeting was adjourned at 8.35 pm.

Table 4 3rd meeting

4th Meeting Minutes

Date : 1st March 2023
Time : 10.30 am
Venue : Common Room IIT
Attendees :

Akash Sathiyanathan
Mohamed Zabrin Mohamed Juzail
Chadithya Dinuwan
Mohammed Alfar
Dilina Chathuraka
Saad Mazhar

Time	Points
10.30 am	○ Meeting was called to order at 10.30 am. ○ Group members submitted their findings with the references to the group leader. ○ Saad Mazhar thanked everyone for their support and mentioned that he will proof read it all and make the necessary changes. ○ Mohamed Juzail asked the group if animations and transitions will be allowed for the presentation. ○ Dilina Chathuraka proposed a template which could be used for the presentation. ○ The group went through every members findings and decided to start working on the presentation as soon as possible. ○ Since there were no more matters to be discussed the meeting was adjourned at 11.30 am.

Table 5 4th meeting

5th Meeting Minutes

Date : 9th March 2023
Time : 1.00 pm
Venue: Common Room IIT
Attendees :

Akash Sathiyanathan
Mohamed Zabrin Mohamed Juzail
Chadithya Dinuwan
Mohammed Alfar
Dilina Chathuraka
Saad Mazhar

Time	Points
1.00 pm	○ Meeting was called to order at 1.00 pm. ○ The report was proofread and compiled by Saad Mazhar. ○ The whole team read the reports and got a clear understanding of the report. ○ The team decided to start working on the presentation immediately. ○ The team discussed if we should hand over the annual reports (which was used for references) as handouts to the panel, to which the majority disagreed to it. ○ Presentation slides were compiled in a creative manner. ○ The presentation script is planned to be discussed in the next meeting. ○ The meeting was adjourned at 3.00 pm.

Table 6 5th meeting

6th Meeting Minutes

Date : 13th March 2023
Time : 4.30 pm
Venue : Common room IIT
Attendees :

Akash Sathiyanathan
Mohamed Zabrin Mohamed Juzail
Chadithya Dinuwan
Mohammed Alfar
Dilina Chathuraka
Saad Mazhar

Time	Points
4.30 pm	○ Meeting was called to order at 4.30 pm. ○ The slides were checked again. ○ The group leader thanked the whole group for their immense support throughout the coursework. ○ The script for the presentation was prepared. ○ The group decided to conduct a few trials runs before the Viva. ○ The meeting was adjourned at 5.00 pm.

Table 7 6th meeting